Pavemind Development/Solution Possibilities

V1.0

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1. Purpose

A customer discussion has highlighted a potential use case for Vehicle Registration Mark (VRM) intelligence within PaveMind, particularly for insurance-related investigations.

The purpose of this brief is to support further customer conversations and requirements discovery without committing to development, delivery timelines or future product capability.

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2. Existing Capability Assessment

2.1 ANPR/OCR Capability

- Do we currently have any capability within our platform to read VRMs from footage, or do we only detect number plate regions for redaction purposes?
- Have we previously evaluated ANPR providers or OCR services that could be leveraged?
- Are there any existing VisionTrack services already extracting vehicle metadata from footage?

2.2 Archive Processing

- Can ANPR/OCR be run retrospectively against footage already stored within our archive?
- What footage retention periods would realistically be available for processing?
- Would reprocessing archived footage require significant infrastructure changes?

3. Prioritise retrospective search

- If a customer supplied a VRM today, what would be the simplest architecture to determine whether that vehicle exists within our footage estate?

Would we:

- Process footage on demand?
- Create a searchable VRM index during ingestion?
- Use a hybrid approach?

3.1 Data Storage

If we extract VRMs, what information would we need to store?

- Registration
- Timestamp
- GPS location
- Device ID
- Confidence score

What would the storage implications be at VisionTrack scale?

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3.2 Search Performance

- How quickly could a VRM search return results?
- Would searches be practical across months of footage or only recent data?

4. Scale & Cost

4.1 Data Volumes

- What is our current daily volume of road-facing footage entering the platform?
- What proportion would be suitable for ANPR processing?
- How many vehicles are likely to be observed daily?

4.2 Infrastructure Impact

- What would be the processing cost per hour of footage?
- Could this be managed using existing infrastructure?
- Would OCR processing become a significant cloud cost driver?

4.3 Commercial Viability

Roughly what level of investment would be required for:

- Proof of Concept
- Pilot
- Production capability

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5. Accuracy & Practical Limitations

5.1 OCR Quality

What accuracy rates should be expected from standard OCR/ANPR solutions?

How would:

- Night footage
- Rain
- Motion blur
- Dirty plates
- Camera angles affect performance?

5.2 False Positives

- How would we manage incorrect reads?
- Would confidence thresholds be required?
- Would customers need to manually validate matches?

6. Operational Model

6.1 Customer Access

- Would users search directly through PaveMind?
- Would searches be performed internally by VisionTrack?
- Should results initially be provided as a managed service rather than a self-service platform?

6.2 Footage Access

Would customers receive:

- A location and timestamp only?
- Associated footage?
- Downloadable evidence packs?

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7. Future Capability

7.1 Live Watchlist Detection

- How different is a live ANPR watchlist solution compared to retrospective search?
- Would detection happen:
  - ✓ In cloud?
  - ✓ At edge/device level?
  - ✓ Via a hybrid model?

7.2 Development Effort

- Is live watchlist functionality a 6-month project or a multi-year roadmap item?
- What additional components would be required?
  - ✓ Real-time processing
  - ✓ Alerting engine
  - ✓ Watchlist management
  - ✓ Audit controls

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